

Cover Letter

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Editor-in-Chief Radiation Physics and Chemistry

Subject: Submission of manuscript for consideration in Radiation Physics and Chemistry

Dear Editor,

I am pleased to submit the manuscript entitled “Effective-Medium Geant4 Predicts K-Edge Nanocomposite Shielding: A Six-Material Explicit-RVE Benchmark” for consideration as an original research article in Radiation Physics and Chemistry.

The manuscript addresses a recurring modelling question in radiation-shielding studies: when experiments describe polymer shields as nanocomposites, is explicit nanoparticle geometry required for macroscopic photon attenuation, or is a composition-correct homogeneous Geant4 material sufficient? To answer this, the study benchmarks six HDPE/high-Z filler systems across three controlled Geant4 regimes: homogeneous effective-medium slabs, 1 μm parameterised explicit representative-volume elements, and 250 nm multi-union explicit representative-volume elements.

The central result is that all six effective-medium materials pass NIST XCOM acceptance criteria, with mean absolute deviations of 0.339-0.793% and maximum deviations of 0.967-2.933%. For the five composition-equivalent binary systems, the 1 μm G4PVPParameterised RVEs agree with matched Regime-A baselines within the pre-registered $\pm 3\%$ H1 gate at 30-150 keV. The multi-union cases reproduce the broad attenuation trends but show larger finite-RVE fluctuations, supporting their use as small-RVE feasibility checks rather than large-RVE production models.

The study also reports an important negative-control result: a ternary HDPE/Bi₂O₃/WO₃ effective medium passes XCOM, but explicit RVEs that spatialise only Bi₂O₃ are not composition-equivalent to the full ternary material and therefore cannot be used to claim H1 validation. This finding gives practical guidance for future multi-filler explicit-RVE studies and prevents overinterpretation of hybrid geometries.

The manuscript is well aligned with Radiation Physics and Chemistry because it combines photon attenuation, shielding-material modelling, Geant4 validation, NIST XCOM comparison, and reproducible computational methodology. Its novelty is methodological rather than material-claim based: it provides a controlled, material-resolved benchmark for deciding when effective-medium modelling is scientifically sufficient and when explicit nanoparticle geometry is warranted.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I confirm that I am the sole author and approve the submitted version. No human participants, animal data, or clinical data were used. No external funding supported this work, and I declare no conflict of interest.

Thank you for considering this manuscript for publication in Radiation Physics and Chemistry.

Sincerely,

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